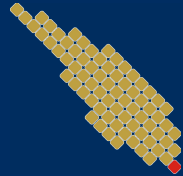




BMKG



ITERA

SOFTWARE OPERATIONAL GUIDEBOOK & USER MANUAL

Department of Geophysical Engineering, Faculty of Industrial Technology

Institut Teknologi Sumatera (ITERA)

SOFTWARE OPERATIONAL TECHNICAL MANUAL

BMKG Strong Motion Analyzer

(BSMA v2.0.0)

Developed at BMKG Sleman Geophysical Station

Computational Platform for Strong Ground Motion Accelerograph Signals & Seismic Kinematics

DOCUMENT METADATA & DEVELOPMENT INFORMATION

Software System Name	: BMKG Strong Motion Analyzer (BSMA)
Release Version / Build	: v2.0.0 (Build 2026.08)
Host Institution	: Sleman Geophysical Station Class I, BMKG D.I. Yogyakarta
Author / Developer	: Ahmad Didane Setyawan Putra
Student ID (NIM)	: 123120094
Study Program	: Geophysical Engineering
Faculty	: Faculty of Industrial Technology
University	: Institut Teknologi Sumatera (ITERA)
Host Partner Institution	: Sleman Geophysical Station Class I, BMKG D.I. Yogyakarta
Internship Period	: July 20, 2026 - August 20, 2026
Project Classification	: Independent Student Academic Work (Non-Commercial)
Web Application URL	: https://strong-motion.streamlit.app/
Source Code Repository	: https://github.com/ahmaddidan/BSMA-v.2

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5-STEP QUICK START OPERATIONAL WORKFLOW

The workflow diagram below summarizes the 5 primary operational steps to execute **BMKG Strong Motion Analyzer (BSMA v2.0.0)** for rapid strong ground motion processing, spanning file loading to engineering deliverables export:

Step 1: Seismic Data Ingestion (Data Ingestion)

Access the cloud web platform or local instance, then upload accelerogram files (.mseed or .sac) in the DATA INPUT sidebar. Upload StationXML (.xml) if records are raw digital counts. Specify recording data provenance.

Step 2: DSP Parameter Configuration (Parameter Settings)

Select output physical units (Gal or m/s^2). Click the 'Standard Strong-Motion Preset (0.05 - 40.0 Hz)' button to instantly apply the standard BMKG-recommended bandpass filter settings.

Step 3: Single-Station Analysis Execution (Run Analysis)

Select the target station from 'Select Recording Window', then click the main '[Run Analysis]' button. The pipeline automatically runs QC screening, 4th-order zero-phase SOS filtering, baseline detrending, and dual kinematic integration.

Step 4: Interactive 7-Tab Dashboard Exploration (Dashboard Exploration)

Navigate interactive tabs to examine kinematic summaries (Tab 1), 3-component time series (Tab 2), QC diagnostics (Tab 3), vibration energy & CAV (Tab 4), MMI intensity (Tab 5), and SNI 1726 structural design response spectra (Tab 6).

Step 5: Deliverables Export & Distribution (Export Deliverables)

On Tab 7, review the print-ready 2-page engineering report preview, then download the official PDF report, time-history CSVs (acc/vel/disp), response spectra CSVs, and unified ZIP archive packages.

[STANDARD PRACTICE] Fast Operational Best Practice

Daily Operational Best Practice: For BMKG operational accelerogram files pre-calibrated by station dataloggers into physical acceleration units (Gal or m/s^2), simply upload the .mseed file, retain default 0.05 - 40.0 Hz bandpass settings, and all analytical tabs and publication-grade PDF reports will be computed automatically in seconds!

CHAPTER I: SYSTEM ACCESS & SEISMIC DATA INGESTION (DATA INPUT)

1.1 Software Interface Access Procedures

BMKG Strong Motion Analyzer (BSMA v2.0.0) can be accessed directly via modern web browsers (Google Chrome, Microsoft Edge, Mozilla Firefox) without complex installation procedures at the public portal: <https://strong-motion.streamlit.app/>. For offline operation on local operational computers within isolated station environments, launch a terminal in the project directory and execute: `streamlit run app.py`. Ensure browser WebGL hardware acceleration is enabled to guarantee high-performance interactive chart rendering.

1.2 DATA INPUT Controls on the Sidebar Panel

- **Accelerogram File Uploader:** Interactive widget to select or drag-and-drop seismic record files. Supported formats include standard FDSN **MiniSEED** (**.mseed**, **.miniseed**, **.msd**) and **SAC** (**.sac**) binaries. The system automatically parses and validates file header structures, extracting network/station codes, component channels (Z, N, E), start times, and sampling rates. BSMA is optimized for records sampled between 20 Hz and 250 Hz.
- **StationXML Uploader (Optional):** Upload zone for sensor response metadata (*.xml). Required only when raw records are uncalibrated digital counts, enabling precise instrument response deconvolution via poles, zeros, and stage gains.
- **'Reset / Clear Data' Button:** Clears the browser session cache, wipes previously loaded data streams, and resets all parameters back to default configurations prior to evaluating new events.

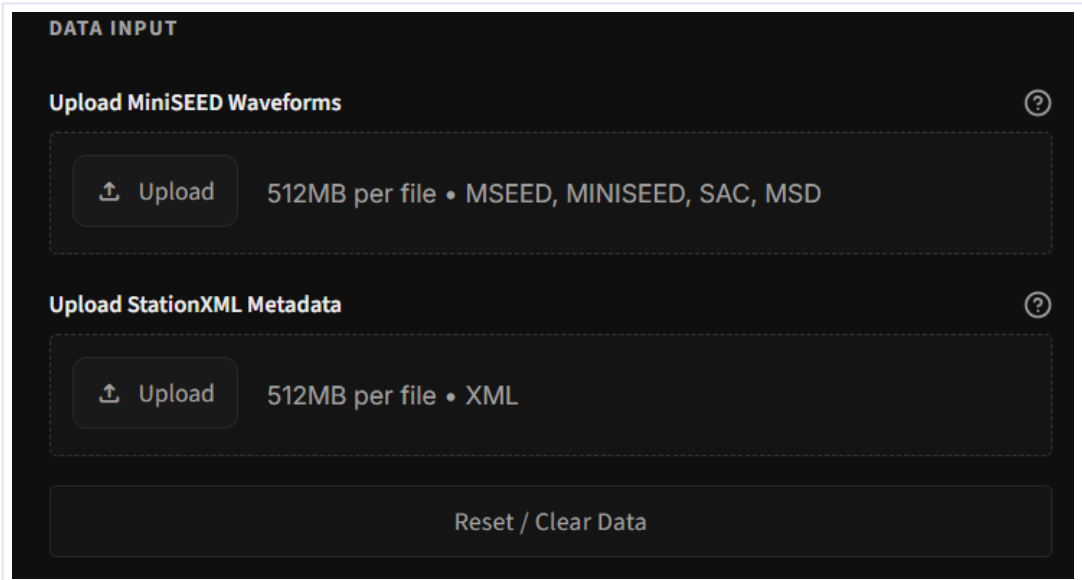


Figure 1.1: Accelerograph File Upload Widget, StationXML Uploader, and Reset Data Button on Sidebar

1.3 Supported Input File Formats in BSMA v2.0.0

Format Type	File Extension	Description & Technical Specifications
MiniSEED	.mseed, .miniseed, .msd	Standard FDSN/IRIS format. Supports triaxial or single-component STEIM-1/STEIM-2 data streams.
SAC Binary	.sac	Seismic Analysis Code format. Stores full seismological headers (station coordinates, sampling delta).
StationXML	.xml	FDSN response metadata containing calibration stages, sensitivity gains, and sensor transfer functions.

[PRACTICAL TIP] Field Seismic Recording Guidelines

Field Recording Prerequisite: Ensure accelerograph files contain at least 5.0 seconds of pre-event quiet window prior to initial P-wave onset. This window is mandatory for BSMA's automated Quality Control (QC) engine to establish the baseline noise floor and compute dynamic Signal-to-Noise Ratios (SNR). Files lacking sufficient quiet windows will trigger QC score penalties.

CHAPTER II: RECORDING ORIGIN DECLARATION (DATA PROVENANCE)

2.1 Scientific Importance of Data Provenance Declarations

In earthquake engineering seismology, accelerograph data distributed from field stations typically exhibit two fundamentally different states: records pre-calibrated into physical acceleration by field dataloggers (e.g., Nanometrics Titan, Guralp, or Kinematics Episensor), and raw records still scaled in digital telemetry units (*counts*). Declaring *Data Provenance* in BSMA v2.0.0 is critical to prevent **accidental double deconvolution**. Applying instrument deconvolution to pre-calibrated physical acceleration signals causes catastrophic high-frequency over-correction, amplifying ambient electronic noise, distorting phase arrivals, and producing false PGA values that invalidate safety analyses.

2.2 Recording Status Options in the PROJECT Sidebar

- **'Already processed physical acceleration'**: Select this option when files are pre-calibrated to physical units (m/s², Gal, cm/s²). The system engages the *calibration bypass protocol*, skipping instrument deconvolution and routing signals directly to zero-phase bandpass filtering and detrending.
- **'Raw instrument counts with StationXML'**: Select this mode when recordings consist of raw integer counts. BSMA ingests the StationXML transfer function and applies frequency-domain response deconvolution stabilized with a water-level regularizer.
- **'Unknown (Require Scientific Review)'**: Selected when the operator is uncertain of the recording's provenance. Heuristic statistical checks are executed against peak amplitudes to recommend appropriate calibration pathways.

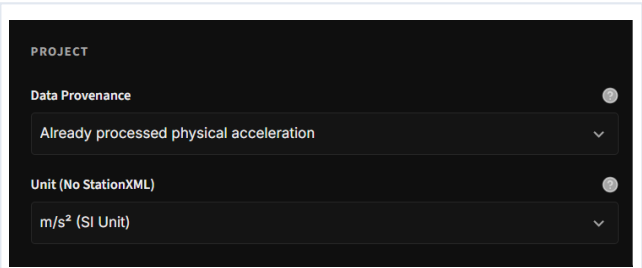


Figure 2.1: Data Provenance Configuration Menu

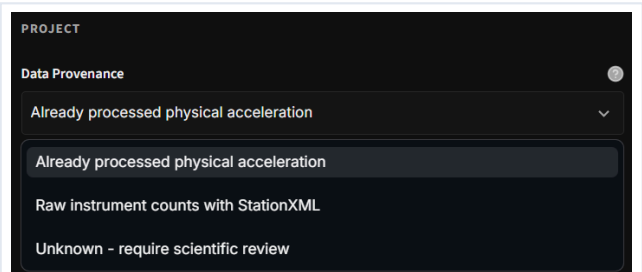


Figure 2.2: Recording Provenance Dropdown Selector

2.3 Recording Provenance Matrix & BSMA Processing Pipelines

Provenance Mode	Typical Amplitude Range	Engine Pipeline Action	Risk of Misconfiguration
Physical Acceleration (Default Recommendation)	Physical floating-point values (0.001 - 500 Gal or m/s ²)	Bypasses deconvolution; direct bandpass & baseline detrending	Selecting for raw counts results in severe PGA underestimation
Raw Counts + XML	Large integer digital values (10 ⁴ - 10 ⁷ digital counts)	Parses StationXML; applies frequency-domain deconvolution	Applying to pre-calibrated signals causes catastrophic over-correction
Unknown Review	Indeterminate provenance	Executes peak amplitude heuristics; flags review warning	Requires visual inspection before confirming baseline integration

[STANDARD PRACTICE] Standard BMKG Signal Calibration Protocol
BMKG Operational Guideline: The vast majority of accelerograph MiniSEED records archived and distributed by BMKG post-earthquake are converted to physical acceleration directly by station field dataloggers. Operators should select 'Already processed physical acceleration' to guarantee kinematic integration stability and response spectra fidelity.

CHAPTER III: DSP PARAMETERS CONFIGURATION (PROCESSING & UNIT)

3.1 Digital Signal Filtering & Conditioning

Digital filtering isolates the engineering-relevant frequency bandwidth of strong ground motions while eliminating instrument noise, thermal drifts, and non-seismic baseline perturbations. BSMA v2.0.0 incorporates standardized DSP workflows:

- **Filter Type Options:** Offers **Bandpass** (standard for strong motion), **Lowpass**, and **Highpass** configurations. Filters are applied using a two-pass *zero-phase forward-backward scheme* based on a 4th-order IIR Butterworth topology in *Second-Order Sections* (SOS) formulation. This ensures zero phase distortion ($\Delta\phi = 0$), preserving exact P-wave arrivals and S-wave kinematics.
- **Corner Frequencies:** Interactive sliders configure the lower corner $f_{\min}$ (default: 0.05 Hz) to suppress low-frequency drift during double integration, and the upper corner $f_{\max}$ (default: 40.0 Hz) to eliminate high-frequency cultural and electrical sensor noise.
- **'Standard Strong-Motion Preset (0.05 - 40.0 Hz)' Shortcut:** A single click restores optimal BMKG/USGS operational corner frequency recommendations.
- **Detrending & Tapering:** *Polynomial Detrending* removes static offsets and thermal polynomial drift, while a 5% *Tukey Cosine Taper* suppresses end-window spectral leakage.

3.2 Physical Engineering Unit Selector

Operators can toggle output units: **m/s²** (SI standard), **Gal** (1 Gal = 1 cm/s² = 0.01 m/s², BMKG operational standard), or **cm/s²**.

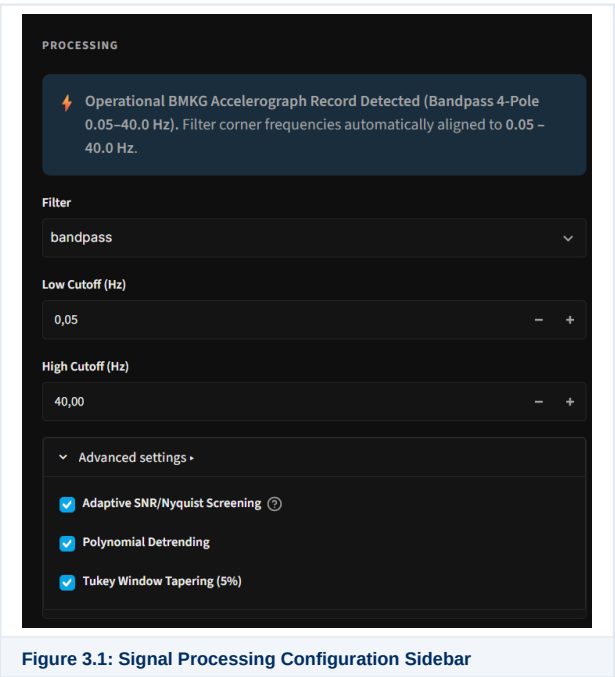


Figure 3.1: Signal Processing Configuration Sidebar

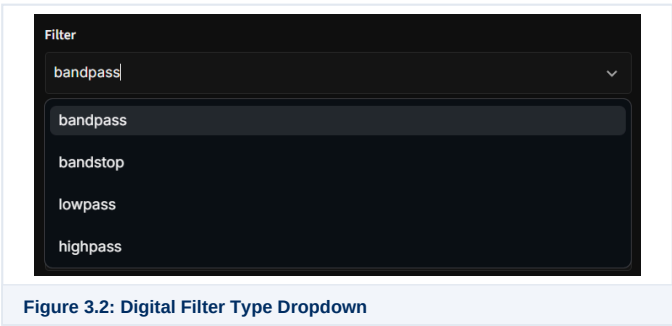


Figure 3.2: Digital Filter Type Dropdown

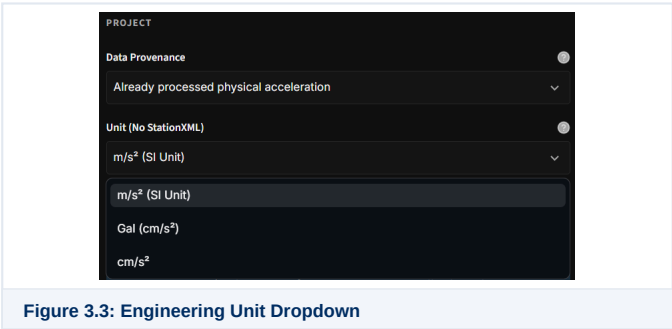


Figure 3.3: Engineering Unit Dropdown

[PRACTICAL TIP] Shannon-Nyquist Frequency Safeguard

Shannon-Nyquist Safeguard: BSMA automatically restricts fmax to a maximum of 0.40 x sampling frequency (fs). This conservative ceiling ensures digital filtering operates safely below the theoretical Nyquist limit (0.50 fs), preventing high-frequency aliasing and filter instability from corrupting engineering data.

CHAPTER IV: STRUCTURAL PARAMETERS, EVENT METADATA, & BENCHMARK

4.1 ANALYSIS Section (SDOF Structural Dynamics & Numerical Solvers)

Single-Degree-of-Freedom (SDOF) elastic oscillator modeling forms the foundation of modern structural earthquake design. BSMA v2.0.0 delivers flexible and rigorous dynamic controls:

- **Critical Damping Ratio (ξ):** Configures structural viscous damping from 0.01 to 0.20. The default value is **0.05 (5%)**, matching normative code provisions for reinforced concrete and structural steel under SNI 1726:2019 and ASCE 7-22.
- **SDOF Solver Algorithms:** Operators can choose between: **Nigam-Jennings (1969)** (exact piece-wise analytical state transition matrix, immune to numerical dispersion and exceptionally accurate at short periods $T < 0.1$ s) or **Newmark-Beta (1959)** (constant average acceleration implicit step-by-step integration with $\gamma=0.5$, $\beta=0.25$, introducing zero artificial numerical damping).

4.2 OUTPUT & BENCHMARK Sections (Event Metadata & Cross-Validation)

- **Event Metadata Form:** Interactive fields for Origin Time (UTC), Magnitude (Mw), Hypocentral Depth (km), Epicentral Coordinates, and Epicentral Distance (km). These parameters populate the official header of the exportable 2-page PDF engineering report.
- **Ground Truth Benchmark:** Upload portal for external reference spectral CSVs, facilitating instant numerical cross-validation.

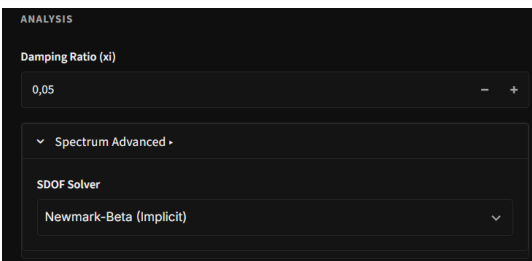


Figure 4.1: Dynamic Analysis Control Sidebar

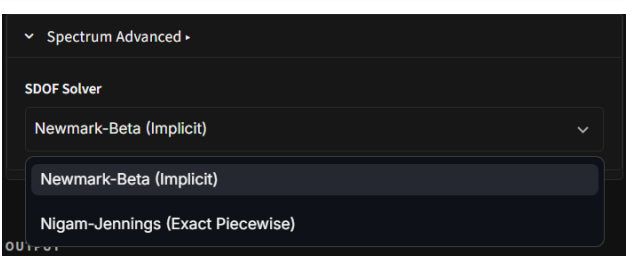


Figure 4.2: SDOF Structural Dynamics Solver Dropdown

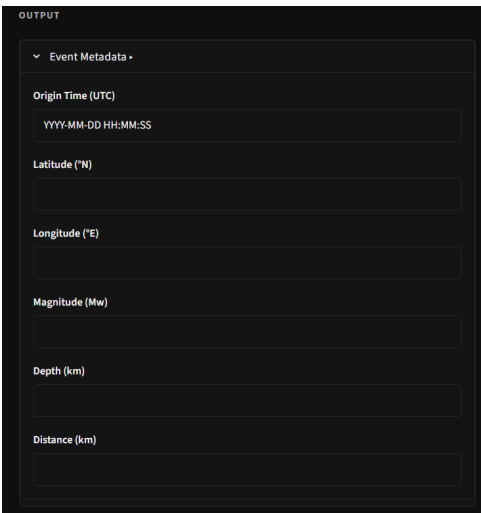


Figure 4.3: Earthquake Event Metadata Form

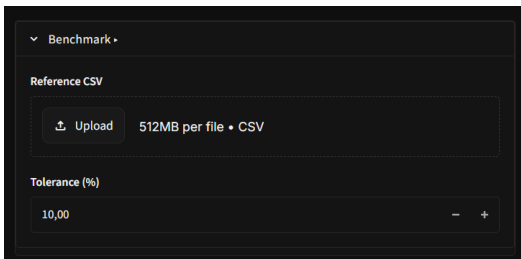


Figure 4.4: Ground Truth Benchmark Upload Widget

[PRACTICAL TIP] Report Metadata Completeness

Engineering Report Completeness: Always complete the earthquake event parameters in 'Output Event Data' prior to downloading the official PDF report in Tab 7. Magnitude, depth, and epicentral distance provide essential seismological context for structural engineers evaluating ground motion attenuation and seismic demand.

CHAPTER V: PIPELINE EXECUTION (RUN ANALYSIS) & TAB 1 SUMMARY

5.1 Workflow Mode Selection & 'Run Analysis' Button Execution

Once sidebar parameters are configured, operators engage the primary workstation area. BSMA v2.0.0 features three main operational modes on the **Workflow Mode** selector: *Single-station review*, *Multi-station processing*, and *Export results*:

- **Single-Station Mode Activation:** Ensure **'Single-station review'** is active for comprehensive station-level signal processing.
- **'Select Recording Window' Dropdown:** Choose the target station stream (e.g., `IA.PPJR | 2026-02-05 18:04:27 UTC`).
- **Calibration Status Feedback:** Displays explicit confirmation, e.g., *Physical Acceleration Mode (StationXML correction bypassed)*.
- **'[Run Analysis]' Execution Button:** Click the primary blue **'Run Analysis'** button to trigger the end-to-end processing pipeline. A dynamic spinner confirms `Executing pipeline for station IA.PPJR...`. The button is automatically disabled if provenance remains 'Unknown' or raw counts lack an accompanying StationXML file.

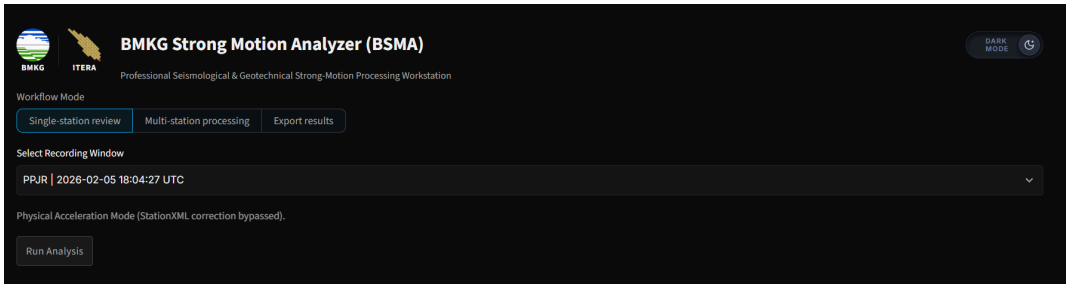


Figure 5.1: Workflow Mode Selector, Station Stream Dropdown, and Primary 'Run Analysis' Execution Button

5.2 'ANALYSIS COMPLETE' Status Badge & Tab 1 SUMMARY Executive Dashboard

Upon pipeline completion, a green **ANALYSIS COMPLETE** badge illuminates in the header, unlocking all 7 interactive tabs simultaneously. The **SUMMARY** tab consolidates station stream metadata (sampling rate, duration, components), the peak kinematic triad (**PGA** [Gal], **PGV** [cm/s], **PGD** [cm]), QC health audit scores, and the instrumental MMI rating.

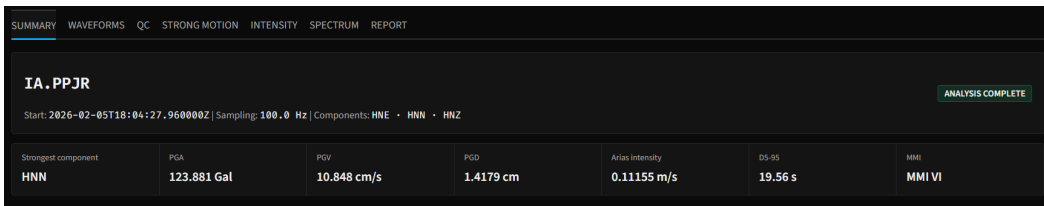


Figure 5.2: 7 Interactive Analysis Tabs Navigation Bar and Station Header with ANALYSIS COMPLETE Badge

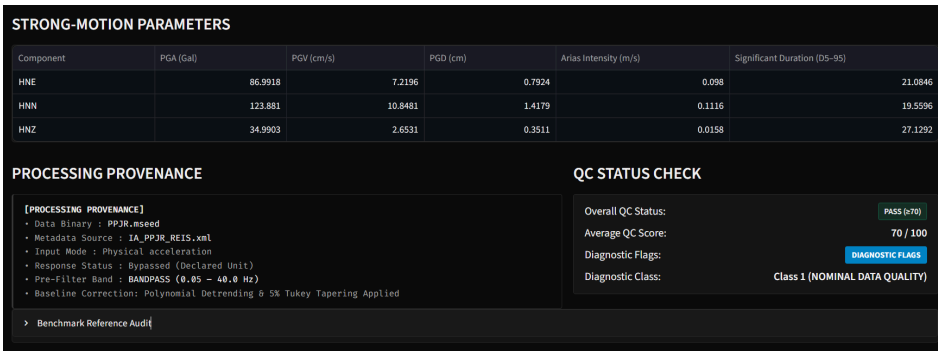


Figure 5.3: Peak Kinematic Parameter Cards (PGA, PGV, PGD), Provenance History, and QC Health Audit

[STANDARD PRACTICE] Rapid Post-Earthquake Verification Protocol

Rapid Post-Earthquake Verification: After triggering 'Run Analysis' and confirming the green badge, immediately review the PGA, PGV, and MMI cards on Tab 1. These metrics provide BMKG duty officers with instant, objective situational awareness of ground shaking severity at the station site prior to detailed waveform inspection.

CHAPTER VI: TAB 2 - FULL KINEMATIC TIME HISTORIES (WAVEFORMS)

6.1 Synchronous Triaxial Kinematic Time Histories

The Waveforms tab provides high-resolution interactive time series resulting from dual kinematic numerical integration:

- **Three-Tier Synchronized Kinematic Rows:** Plots **Acceleration $a(t)$** [Gal], **Velocity $v(t)$** [cm/s], and **Displacement $d(t)$** [cm] stacked along a shared horizontal time axis. This layout facilitates direct physical correlation between high-frequency acceleration pulses, intermediate velocity energy bursts, and long-period displacement offsets.
- **Interactive Component Channel Selector:** Allows isolating individual orthogonal channels (East-West HNE, North-South HNN, or Vertical HNZ) or displaying all three components simultaneously for comprehensive 3D motion characterization.

6.2 Plotly WebGL Interactivity & Automated Peak Annotations

- **Wave Phase Inspection:** Supports seamless box zoom on initial P-wave and high-energy S-wave arrivals, horizontal panning, and precision cursor hover tooltips displaying instantaneous amplitudes and timestamps.
- **Automated Peak Annotations:** Prominent vertical dashed lines and dynamic text tags mark the exact occurrence of PGA, PGV, and PGD.



Figure 6.1: Synchronized Kinematic Acceleration, Velocity, and Displacement Waveform Time Histories

6.3 Physical Significance of Ground Kinematics in Earthquake Engineering

- **Ground Acceleration $a(t)$:** Directly governs instantaneous inertial shear forces ($F = m \cdot a$) transmitted to stiff low-rise structures.
- **Ground Velocity $v(t)$:** Reflects kinetic energy flux and governs demand in mid-rise structures and buried lifeline networks (pipelines).
- **Ground Displacement $d(t)$:** Controls transient inter-story drift demands in flexible high-rises and permanent ground strain.
- **Integration Drift Check:** If displacement tails drift non-linearly away from zero, raise f_{min} (e.g., 0.05 Hz $\rightarrow$ 0.10 Hz) in the sidebar.

[OPERATIONAL ADVISORY] Displacement Baseline Stability Verification
Baseline Stability Diagnostics: Closely examine the displacement trace toward the end of the recording. If the displacement waveform drifts upwards or downwards away from zero, increase f_{min} in the sidebar (e.g., from 0.05 Hz to 0.10 Hz) and ensure Polynomial Detrending is enabled to suppress residual low-frequency integration noise.

CHAPTER VII: TAB 3 - SIGNAL QUALITY CONTROL GATES (QC GATES)

7.1 Automated Signal Quality Auditing (QC Diagnostic Engine)

The Quality Control tab provides an objective multi-gate audit of raw accelerograms prior to downstream kinematic integration. A composite score is calculated from a 100-point base: $QC\ Score = \max(0, \min(100, 100 - \sum Penalties))$ across 6 physical gates:

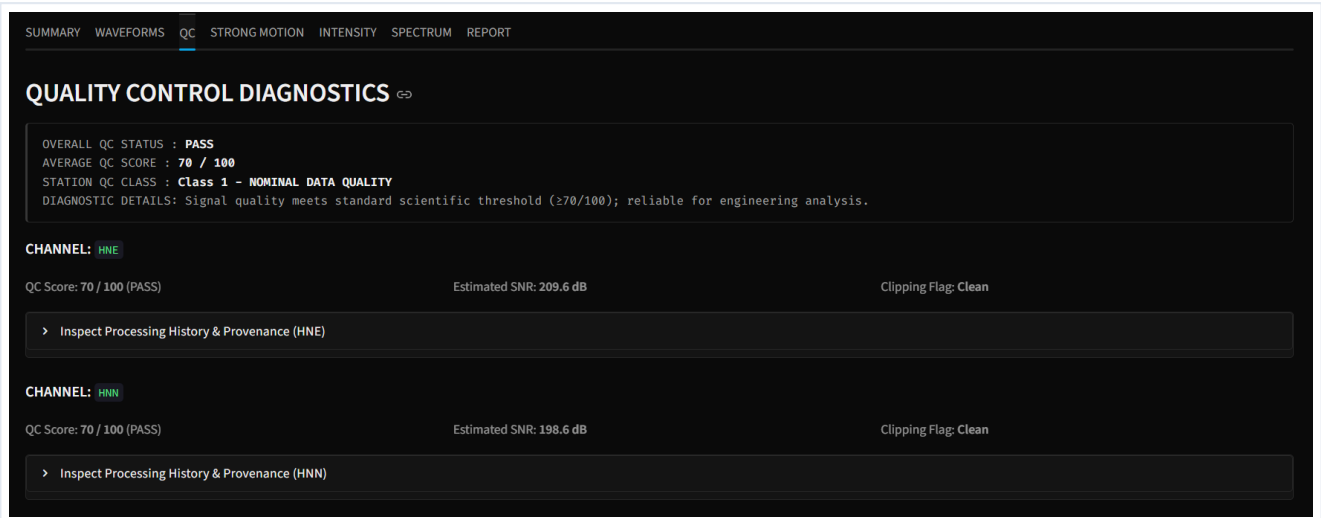


Figure 7.1: Signal Quality Control Dashboard (QC Health Score, Pre-Event SNR, and 6 Physical QC Gates)

7.2 Six Physical Diagnostic Gates (QC Gates) & Penalty Structure

Severity Tier	Physical Anomaly Criteria & Gate Parameters	Penalty Deduction	Engineering Eligibility
Nominal / Pristine	Clean waveform, low noise floor, pre-event SNR ≥ 10 dB, stable baseline	0 (Base 100)	PASS (≥ 70 Points) Fully approved for analysis
Warning Tier (Advisory)	- Static baseline offset (> 2% PGA) - Thermal baseline drift (> 5%) - Pre-event window < 5.0 s - Marginal pre-event SNR (3 to 10 dB)	-15 Points (per anomaly)	WARNING (50-69 Points) Acceptable with engineering caveats
Major Degradation (Severe Noise)	- Severe ambient noise (SNR < 3 dB) - Multiple high-amplitude spikes (MAD ≥ 6.0)	-40 Points (per anomaly)	Quality degradation zone
Critical Anomaly (Fatal Override)	- ADC saturation (<i>clipping</i> ≥ 0.02 s / 98% FS) - Channel dropout (<i>flatline</i> ≥ 1.0 s) / corrupted file	Fatal Override (Score = 0)	FAIL / REJECT (< 50 Points) Rejected from integration

[PRACTICAL TIP] Interpretation of Field Record Score 70

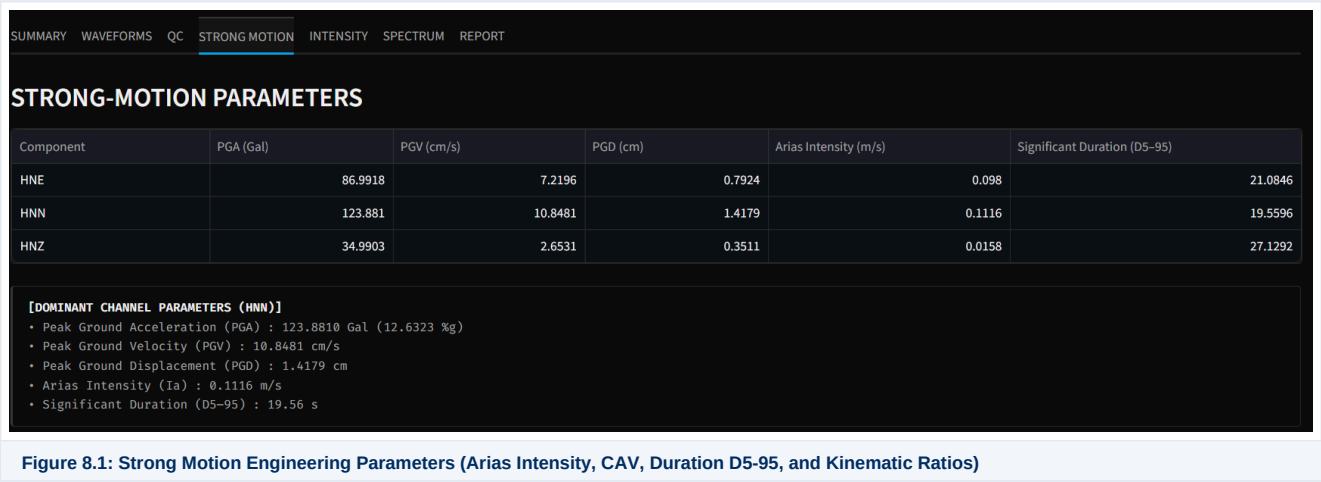
Interpreting Score 70 on Natural Field Records: A score of 70 denotes a fully valid raw recording that simply carries (1) Baseline Offset (-15 Points) and (2) Natural Thermal Drift (-15 Points). Score: $100 - 30 = 70$ (PASS status). Both non-fatal anomalies are completely eliminated by BSMA's polynomial detrending and zero-phase Butterworth filter!

CHAPTER VIII: TAB 4 - STRONG GROUND MOTION & SEISMIC ENERGY

8.1 Earthquake Engineering Energy Parameter Characterization

The Strong Motion tab quantifies engineering parameters reflecting destructive energy potential, ground motion duration, and cumulative dynamic demands imposed on superstructure and geotechnical foundations:

- **Arias Intensity (Ia):** Total seismic energy flux derived by integrating squared acceleration across record duration: $I_a = (\pi / 2g) \int a^2(t) dt$ (units: m/s). This parameter is vital for slope stability assessments (Newmark sliding block), liquefaction triggering evaluations, and residual deformation modeling in dams and embankments.
- **Cumulative Absolute Velocity (CAV):** Time integral of absolute acceleration: $CAV = \int |a(t)| dt$ (units: g-s). Under EPRI (1988) criteria, thresholds of $CAV \geq 0.16 \text{ g-s}$ combined with $PGA \geq 0.10 \text{ g}$ serve as the benchmark standard mandating structural safety inspections in nuclear power stations and strategic infrastructure.
- **Trifunac-Brady Significant Duration (D5-95):** The time interval spanning 5% to 95% total Arias Intensity accumulation. It represents the strong shaking duration responsible for cyclic stiffness degradation and structural collapse mechanisms.
- **Kinematic Ratio Vmax / Amax:** Provides insight into dominant site ground periods ($T_s \approx 2\pi V_{max}/A_{max}$) and local amplification due to soft sedimentary cover.



8.2 Strong Motion Parameters & Structural Damage Correlations

Seismic Parameter	Unit	Civil Engineering Application & Structural Impact
Arias Intensity (Ia)	m/s	Liquefaction triggering in saturated sands and seismic slope displacement evaluations.
Cumulative Abs. Vel. (CAV)	g-s	Damage screening for stiff structures (EPRI 1988 threshold: $CAV \geq 0.16 \text{ g-s}$).
Significant Duration (D5-95)	seconds	Governs cyclic loading demand and progressive structural fatigue degradation.
Kinematic Ratio Vmax / Amax	seconds	Estimates dominant site period ($T_s \approx 2\pi V_{max}/A_{max}$); flags sedimentary basin resonance.

[STANDARD PRACTICE] EPRI 1988 Damage Screening Standard

EPRI Damage Screening Criteria: When recorded CAV exceeds 0.16 g-s and PGA surpasses 0.10 g, critical civil infrastructure including bridges, dams, industrial plants, and multi-story structures should undergo visual structural safety inspections to detect micro-fatigue cracking or differential foundation settlements.

CHAPTER IX: TAB 5 - INSTRUMENTAL INTENSITY ESTIMATION (INTENSITY MMI)

9.1 Instrumental Modified Mercalli Intensity (MMI) Estimation

The Intensity tab translates recorded ground kinematics into objective instrumental intensity levels using peer-reviewed Ground-Motion to Intensity Conversion Equations (GMICE):

- **GMICE Formulations (Worden et al., 2012):** Standard operational relationship utilized by USGS ShakeMap. MMI is derived from the maximum horizontal peak amplitude (**Max-H** across HNE and HNN channels, excluding the vertical HNZ component).
- **Horizontal Max-H Rationale:** Vertical motions are deliberately excluded because civil structures are designed with substantial static gravity margins; severe damage and collapse are predominantly triggered by lateral horizontal inertial shear demands.
- **Kinematic Crossover Logic:** PGA governs intensity estimates during weak shaking ($MMI < 5.0$). Conversely, PGV governs moderate to destructive intensity ranges ($MMI \geq 5.0$), where structural damage correlates with kinetic energy and velocity pulse deformations.

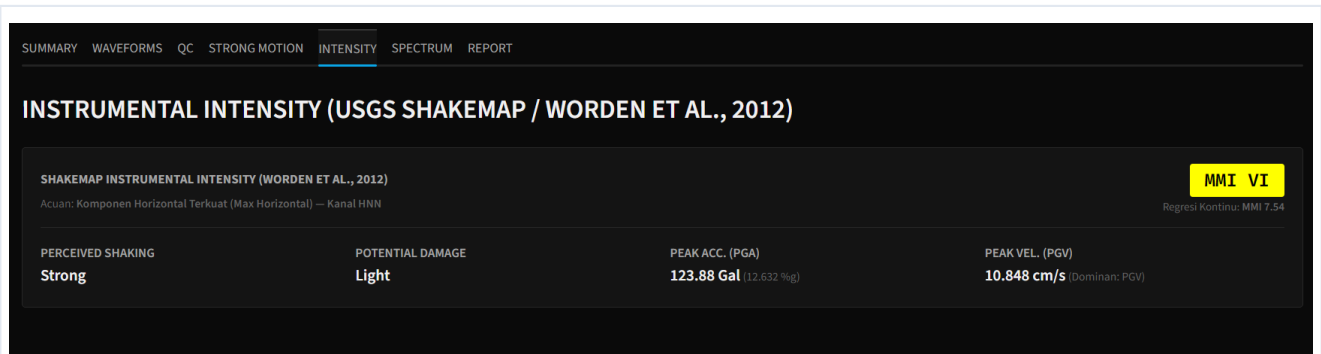


Figure 9.1: Max-H Instrumental MMI Intensity Scale Display and Parameter Summary on Tab 5

9.2 USGS ShakeMap Standard MMI Classification Matrix (Worden et al., 2011/2012)

The standard reference matrix below details quantitative correlations across Perceived Shaking, Potential Damage, Peak Acceleration **PGA** (%g), Peak Velocity **PGV** (cm/s), and **Instrumental Intensity** (MMI I - X+):

PERCEIVED SHAKING	Not felt	Weak	Light	Moderate	Strong	Very strong	Severe	Violent	Extreme
POTENTIAL DAMAGE	none	none	none	Very light	Light	Moderate	Mod./Heavy	Heavy	Very Heavy
PEAK ACC.(%g)	<0.05	0.3	2.8	6.2	12	22	40	75	>139
PEAK VEL.(cm/s)	<0.02	0.1	1.4	4.7	9.6	20	41	86	>178
INSTRUMENTAL INTENSITY	I	II-III	IV	V	VI	VII	VIII	IX	X+

Scale based upon Worden et al. (2011)

Figure 9.2: USGS ShakeMap Instrumental Intensity (MMI) Quantitative Relationship Matrix (Worden et al., 2011)

[PRACTICAL TIP] Instrumental MMI Operational Boundaries & Unit Equivalencies

Instrumental MMI Operational Boundaries: Tab 5 intensity ratings represent objective point measurements derived from single station instruments. While critical for rapid automated situational awareness and immediate post-earthquake briefing, instrumental MMI does not replace comprehensive post-event field macroseismic surveys accounting for building typology variations and micro-geological site effects. Unit equivalence note: 1 %g equals approximately 9.81 Gal.

CHAPTER X: TAB 6 - SDOF RESPONSE SPECTRA & SNI 1726 CODE OVERLAYS

10.1 SDOF Elastic Response Spectra & SNI 1726:2019 Design Code Overlays

The Spectrum tab models the response of 5% critically damped ($\xi = 0.05$) linear Single-Degree-of-Freedom systems across natural periods $T = 0.01$ s to 10.0 s, identifying resonance vulnerabilities:

- **Pseudo-Spectral Acceleration (PSA) Curves:** Quantifies peak spectral acceleration acting on structures as a function of their fundamental vibration periods. Horizontal axes toggle between Linear (for short-period resolution) and Logarithmic scales.
- **SNI 1726:2019 Design Spectrum Overlays:** Compares actual recorded response spectra against official Indonesian Building Code design envelopes for **Soft Soil (SE)**, **Medium Soil (SD)**, and **Hard Soil (SC)**, revealing whether seismic ground demands exceeded design capacity.
- **SDOF Solver Cross-Validation Panel:** Direct comparative plotting of Nigam-Jennings (1969) analytical vs Newmark-Beta (1959) numerical spectra, demonstrating near-zero numerical divergence across all evaluated periods.

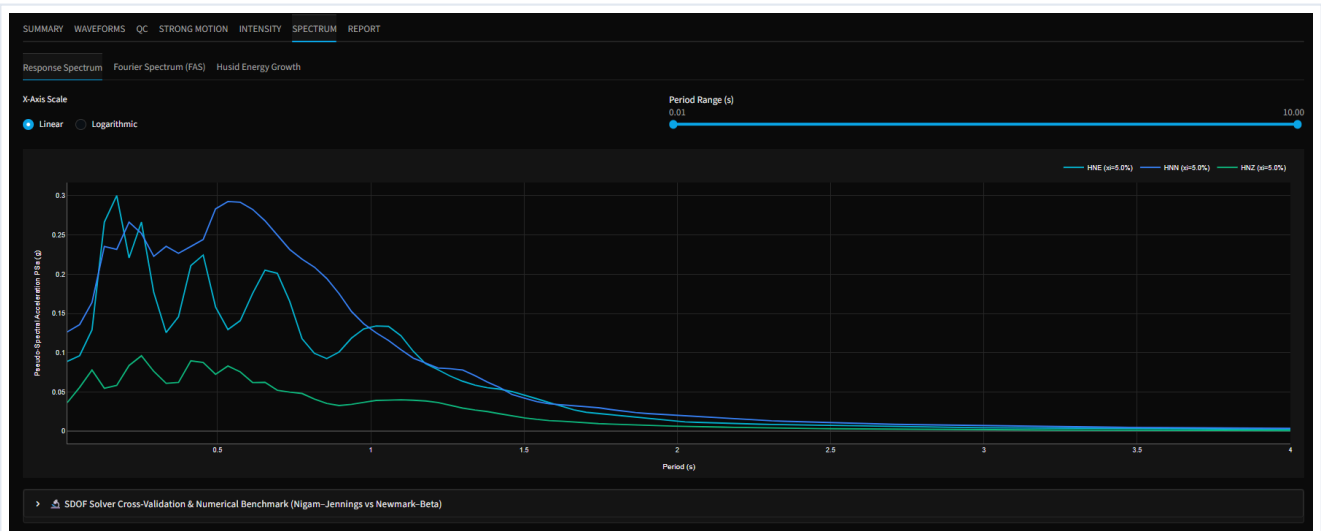


Figure 10.1: 5% Damped Elastic Response Spectra PSA Compared with SNI 1726:2019 Code Design Envelopes

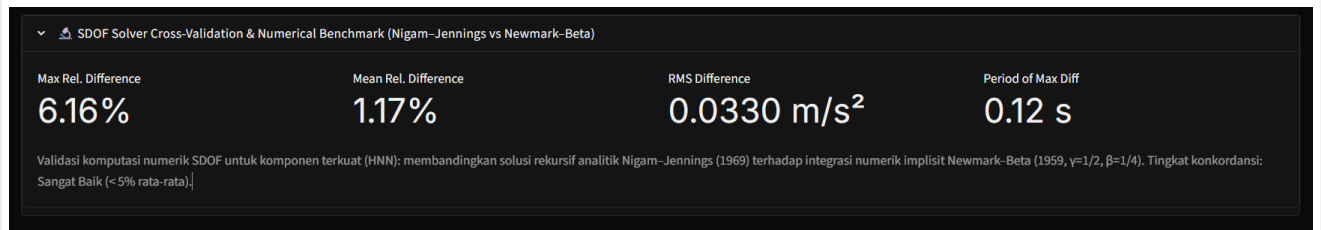


Figure 10.2: SDOF Solver Numerical Cross-Validation Panel (Nigam-Jennings vs Newmark-Beta)

[PRACTICAL TIP] Structural Resonance Vulnerability Identification

Structural Resonance Detection: Identify the dominant period where the PSA spectrum peaks. If peak amplification concentrates between $T = 0.1 - 0.4$ s, low-to-medium rise buildings (1-4 stories) face the highest dynamic resonance risk. Conversely, peak spectral demands at $T > 1.0$ s threaten flexible high-rises and long-span bridges.

CHAPTER XI: ADVANCED SPECTRAL ANALYSIS (FAS SPECTRUM & HUSID PLOT)

11.1 Fourier Amplitude Spectrum (FAS) & Dominant Site Frequencies

The Fourier Amplitude Spectrum transforms acceleration time series into the continuous frequency domain (Hz) via Fast Fourier Transform (FFT). Unlike SDOF response spectra, FAS deconstructs pure harmonic energy distributions uncoupled from structural damping:

- **Dominant Site Frequency (f_0):** Peak spectral amplitudes identify fundamental soil resonance frequencies, governed by sedimentary overburden thickness and shear wave velocity (V_s) contrasts with underlying engineering bedrock.
- **Soil-Structure Resonance:** When site dominant frequencies coincide with the fundamental frequency of the overlying building, *double resonance* phenomena can occur, amplifying structural base shear and catastrophic damage.

11.2 Seismic Energy Accumulation (Husid Energy Growth Plot)

The Husid plot portrays the cumulative time-rate of squared acceleration energy normalized from 0% to 100%:

- **Profile Morphology:** Steep vertical slopes indicate impulsive, concentrated energy release (near-fault forward directivity / fling-step). Gradual, extended rises characterize prolonged energy dissipation common in large megathrust subduction earthquakes.
- **Significant Duration Calibration:** Horizontal markers at 5% and 95% threshold lines calibrate Trifunac-Brady duration (D_{5-95}).

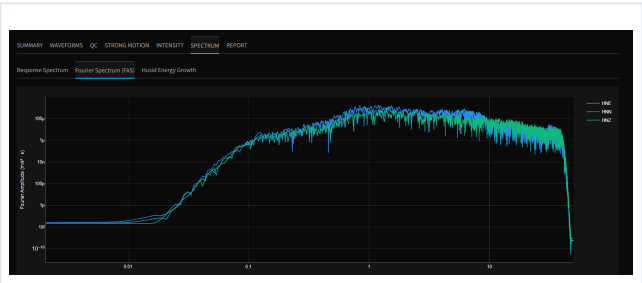


Figure 11.1: Fourier Amplitude Spectrum (FAS)



Figure 11.2: Cumulative Husid Seismic Energy Growth Curve

11.3 Earthquake Source Rupture Mechanisms vs Husid Curve Morphology

Rupture Mechanism	Husid Profile Morphology	Duration D_{5-95}	Structural Engineering Implications
Near-Fault Directivity	Abrupt vertical step within initial seconds	Short (< 10 seconds)	High-velocity pulse loading; severe plastic deformation demands
Shallow Crustal	Uniform moderate-slope monotonic rise	Moderate (10 - 25 s)	Symmetric cyclic reversals; progressive structural degradation
Megathrust Subduction	Extended multi-phase gradual rise	Long (> 35 seconds)	High-cycle fatigue damage; severe soil liquefaction triggering

[STANDARD PRACTICE] Geotechnical Engineering Applications of Spectral Analysis

Geotechnical & Structural Synergy: Combining FAS frequency peaks with PSA design spectra yields comprehensive insight into ground motion site effects. Engineers can pinpoint potential resonance risks, guide seismic retrofitting prioritization, and calibrate foundation damping requirements.

CHAPTER XII: TAB 7 - PDF ENGINEERING REPORT EXPORT & DELIVERABLES

12.1 Engineering Seismological Deliverables Export Panel

The Report tab provides a unified export suite generating standardized, publication-ready engineering outputs:

- **2-Page Official Engineering PDF Report:** Generates a high-resolution vector PDF report (via PyMuPDF) containing BMKG-ITERA logos, event metadata, PGA/PGV/PGD KPI cards, synchronized 3-component kinematic traces, Husid curves, PSA vs SNI 1726 overlays, and QC audit tables.
- **Corrected Kinematic Time History CSVs:** Exports discrete acceleration, velocity, and displacement series for dynamic time-history analyses in structural engineering software (ETABS, SAP2000, OpenSees).
- **Response Spectra CSVs:** Tabulates natural period T against PSA, PSV, and SD spectral ordinates for design code verification.
- **Unified ZIP Archive Package:** Bundles all output files (PDF reports, CSV tables, JSON provenance logs) into a single archive.

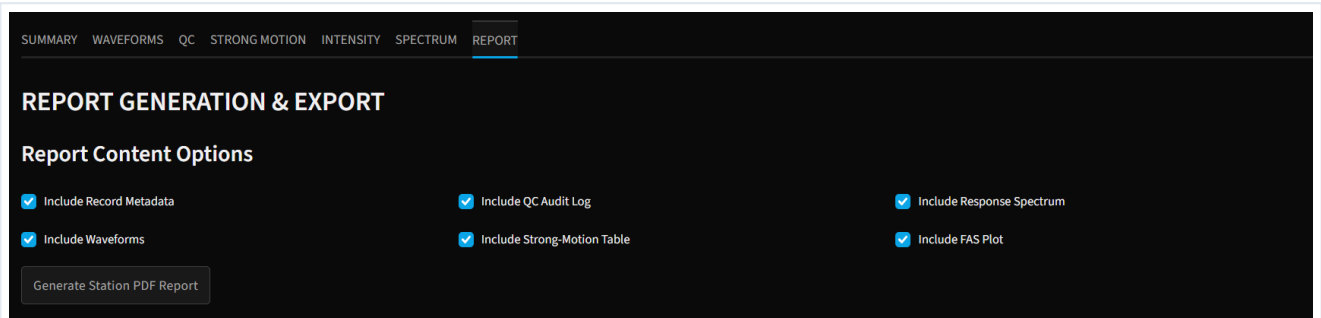


Figure 12.1: Interactive Preview Panel of the Official 2-Page Engineering Seismology PDF Report

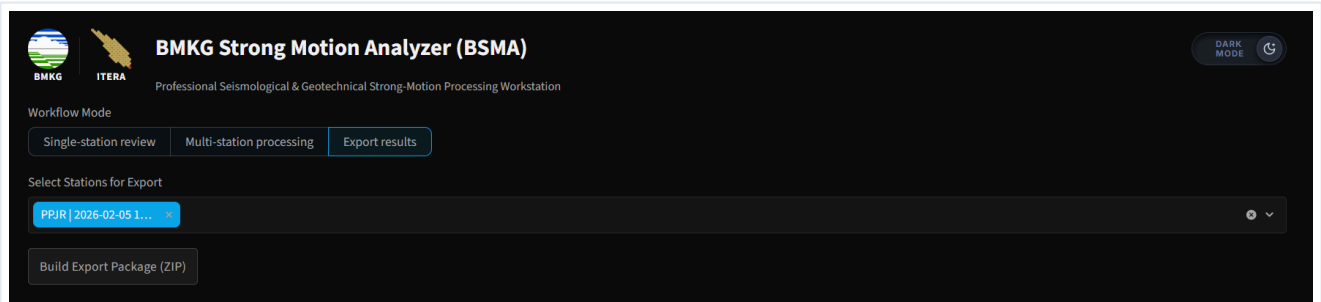


Figure 12.2: Deliverables Download Widgets (PDF Report, Kinematic CSV, Spectral CSV, JSON, and ZIP Archive)

12.2 BSMA v2.0.0 Engineering Deliverables Format Summary

Deliverable File Name	File Format	Primary Engineering Use & System Integration
Engineering PDF Report	Vector PDF (*.pdf)	Official post-event stakeholder briefings, BMKG archives, and technical reports.
Corrected Kinematic CSV	Spreadsheet CSV (*.csv)	Non-linear dynamic time-history analysis in structural software (ETABS, OpenSees).
Response Spectra CSV	Spreadsheet CSV (*.csv)	Equivalent lateral force and spectral modal analysis for building design.
Unified Archive Package	Compressed ZIP (*.zip)	Permanent institutional record archiving complete analytical data packages.

CHAPTER XIII: REGIONAL MULTI-STATION BATCH PROCESSING PROTOCOL

13.1 Regional Seismic Network Processing Rationale

Significant earthquake events trigger ground motion recordings across dozens of regional BMKG accelerograph stations simultaneously. Processing each station stream sequentially by hand is inefficient during emergency crisis operations. **Multi-Station Batch Processing Mode** in BSMA v2.0.0 automates bulk folder ingestion, parallel screening, and regional summary matrix synthesis.

13.2 Batch Workflow Execution Steps

- **Batch Mode Selection:** Switch to **'Batch Processing Mode'** on the top navigation bar.
- **Bulk File Upload:** Drag-and-drop a collection of MiniSEED (.mseed) or SAC (.sac) files from regional stations into the bulk uploader.
- **Fault Isolation Architecture:** Each station stream executes within an isolated boundary. Corrupted traces are flagged as **FAILED** without aborting or stalling batch processing for remaining stations.
- **Regional Comparison Matrix:** Compiles all processed station metrics: PGA, PGV, PGD, Arias Ia, D5-95, MMI, and QC Health Scores.
- **'Export Regional Summary CSV' Button:** Downloads the synthesized regional matrix for immediate GIS mapping (QGIS/ArcGIS).

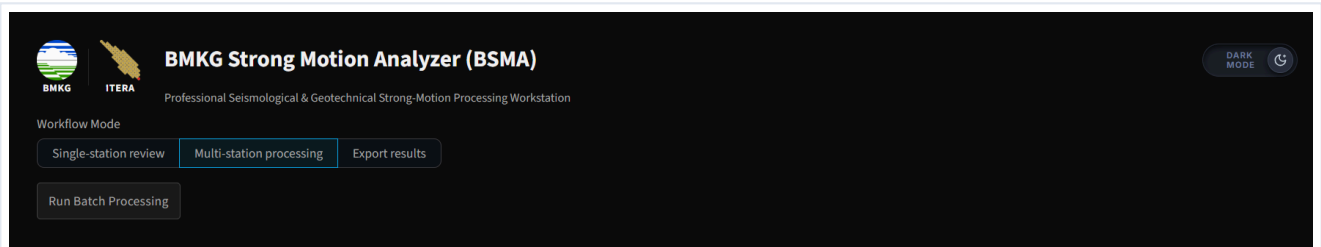


Figure 13.1: Regional Multi-Station Batch Processing Dashboard and Seismic Network Comparison Matrix

13.3 Regional Summary CSV Column Structure & GIS Software Integration

CSV Data Column	Measurement Unit	Spatial Mapping & GIS Analysis Application
Station, Net, Chan	Identity String	Unique primary key for joining station geospatial coordinate layers.
PGA_MaxH, PGV_MaxH	Gal, cm/s	Core variables for spatial shaking contour interpolation (Kriging / IDW).
Arias_Ia, CAV	m/s, g-s	Geotechnical hazard overlays for liquefaction and slope instability mapping.
MMI_MaxH, QC_Score	Numerical Scale, Points	Color symbology for instrumental ShakeMaps and quality assurance filtering.

[STANDARD PRACTICE] Emergency Disaster Response Utilization

Disaster Response Acceleration: During emergency post-earthquake operations, Batch Mode allows duty seismologists to compile comprehensive PGA, PGV, and MMI ShakeMap distributions across entire regional networks within seconds, dramatically accelerating critical situation briefings for civil protection agencies (BPBD/BNPB).

CHAPTER XIV: OPERATIONAL TROUBLESHOOTING GUIDE & FIELD PROCEDURES

The table below summarizes common technical anomalies encountered during software operation along with standardized operational resolution workflows:

Observed Symptom	Probable Root Cause	Operational Solution & Corrective Action
File ingestion failure / rejected format	Corrupted file header, non-standard extension, or temporal telemetry gaps.	Verify file is standard FDSN MiniSEED (*.mseed) or SAC (*.sac) without sample discontinuities.
Displacement time history drifts upward/downward	Residual long-period noise amplified during numerical double integration.	Increase lower corner f_{min} (e.g., 0.05 → 0.10 Hz) and ensure Polynomial Detrending is enabled.
Unreasonable amplitude surge (10 ⁶ Gal)	Instrument deconvolution erroneously applied to pre-calibrated physical data.	Set Data Provenance to ' <i>Already processed physical acceleration</i> ' to engage calibration bypass.
QC status flags WARNING (Score 70)	Natural static baseline offset or moderate thermal drift in raw field sensor.	Normal status for field records. BSMA bandpass filtering and detrending cleanse anomalies automatically.
QC status flags FAIL / REJECT (Score 0)	ADC saturation (sensor clipping) or channel dead flatline.	Physically corrupted record; data cannot be integrated into velocity/displacement validly.
PDF report download fails	Session cache exhaustion or incomplete pipeline execution.	Click 'Reset / Clear Data' on sidebar, reload record, and ensure pipeline execution completes fully.
Batch mode stalls on single station	Corrupted file lacking mandatory FDSN headers.	BSMA fault isolation automatically skips the invalid record and continues processing remaining stations.

14.2 System Operational Boundaries & Computational Constraints

- **Channel Integrity:** Requires standard triaxial orthogonal components (Z, N, E). If single-channel records are loaded, Max-H MMI is calculated from the available horizontal component.
- **Sampling Rate Uniformity:** Supports uniform rates between 20 Hz and 250 Hz. Streams with variable sampling intervals must be resampled before loading.
- **Memory Recommendations:** For extensive batch directories (> 50 stations), a machine with ≥ 8 GB RAM is recommended to ensure fluid buffer streaming and chart rendering.

[PRACTICAL TIP] User Technical Support & Issue Reporting

Technical Support & Issue Tracking: If operational difficulties persist after applying the troubleshooting steps above, contact the developer via official university email or submit a detailed issue report on the official BSMA GitHub repository.

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[PRACTICAL TIP] Academic Attribution & Open-Source Statement

Academic Attribution & Open-Source Statement: BMKG Strong Motion Analyzer (BSMA v2.0.0) and this operational technical manual constitute original academic works developed by an undergraduate Geophysical Engineering student of ITERA during an internship at BMKG Sleman. All source code is open-source to support advancements in engineering seismology and earthquake disaster mitigation in Indonesia.